

Cloud Shield - AI

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Github Url: <https://github.com/jyoshana-deviii/cloudshield-ai>

Description:

Cloud Shield AI is an AI-powered cloud security and monitoring platform designed to protect AWS environments from unauthorized access, security threats, and resource misuse. The system continuously monitors cloud activities in real-time, detects suspicious behaviour using intelligent analysis, sends instant alerts to administrators, and automates security responses to minimize risks. It also provides centralized visibility for multi-service AWS environments, helping organizations improve cloud security, compliance, and operational efficiency through a single unified dashboard.

Functional Requirements

Authentication & Access Control

- ◆ Secure user registration, login, and logout
- ◆ Multi-factor authentication (MFA) support
- ◆ Role-based access control (Admin, Security Analyst, Viewer)
- ◆ Secure password encryption and token refresh mechanism

AWS Environment Integration

- ◆ Connect and monitor multiple AWS accounts
- ◆ Support for multi-region cloud environments
- ◆ Secure AWS IAM role/API integration
- ◆ Add, update, and remove AWS accounts from the dashboard

Threat Detection & Monitoring

- ◆ Real-time monitoring of AWS activities and login events
- ◆ Detect suspicious logins, unauthorized access, and abnormal behavior
- ◆ AI-based anomaly detection for cloud activities
- ◆ Continuous monitoring of EC2, S3, IAM, Lambda, RDS, and CloudTrail

Security Intelligence

- ◆ Automatic detection of security vulnerabilities and misconfigurations
- ◆ Identify publicly exposed resources and risky IAM permissions
- ◆ Threat severity classification (Low, Medium, High, Critical)
- ◆ Security risk scoring for AWS accounts

Automated Incident Response

- ◆ Trigger automatic alerts during suspicious activities
- ◆ Auto-disable compromised access keys or suspicious sessions
- ◆ Automated resource isolation for detected threats
- ◆ Real-time incident tracking and response logs

Alerts & Notifications

- ◆ Instant email and in-app security notifications
- ◆ Alerts for unauthorized login attempts and unusual activity
- ◆ Custom alert rules for security events and resource behavior
- ◆ Notification history and alert management

Resource & Usage Analysis

- ◆ Scan and display active AWS resources
- ◆ Identify unused, idle, or suspicious resources
- ◆ Resource filtering by service, region, and account
- ◆ Real-time cloud utilization insights

Activity Logs & Audit Tracking

- ◆ CloudTrail integration for live activity monitoring
- ◆ Maintain audit logs for user and system actions
- ◆ Search and filter logs by user, service, or timestamp
- ◆ Generate forensic investigation reports

Team & User Management

- ◆ Invite and manage team members securely
- ◆ Assign security roles and permissions
- ◆ Monitor member activities and access history
- ◆ Remove or restrict unauthorized users

Reports & Dashboard

- ◆ Unified dashboard for cloud security overview
- ◆ Generate monthly security and threat reports
- ◆ Export reports in PDF and CSV formats
- ◆ Shareable read-only dashboard/report access

AI-Powered Recommendations

- ◆ AI-generated security improvement suggestions
- ◆ Predict potential threats using behaviour analysis
- ◆ Recommend IAM policy hardening and best practices
- ◆ Cloud security optimization insights based on AWS standards

Non-Functional Requirements

- ◆ Dashboard page load time should be under 3 seconds
- ◆ Real-time threat detection alerts should be generated within 5 seconds
- ◆ API response time should remain under 2-3 seconds under normal load
- ◆ System should support 100+ concurrent users simultaneously
- ◆ Target system availability should be 99.9% uptime

Problem Statement

Organizations using AWS cloud environments often struggle to monitor security threats, unauthorized access, and suspicious activities across multiple cloud services and accounts. Existing cloud security solutions are often expensive, complex to configure, and difficult for small teams or educational projects to manage effectively. Engineering and DevOps teams need a lightweight, AI-powered security platform that can continuously monitor AWS resources, detect potential cyber threats in real time, automate security alerts and responses, and provide a centralized dashboard for cloud protection without heavy infrastructure or high enterprise costs.

Proposed Solution

CloudShield AI is an AI-powered AWS cloud security monitoring system designed to detect suspicious login activities and protect AWS accounts in real time. The system collects activity logs from AWS CloudTrail and GuardDuty, processes them using AWS Lambda and Amazon Bedrock AI models, analyzes user behaviour patterns, and determines whether activities are normal or suspicious. Based on the AI decision, the platform sends instant alerts through SNS, updates the security dashboard, and stores activity logs securely in Amazon S3 for auditing and future analysis.

Technologies Used

Amazon CloudTrail, Amazon GuardDuty, Lambda, Bedrock, SNS, S3.

System Architecture

CloudShield AI follows a three-layer architecture. AWS CloudTrail logs, GuardDuty findings, and user behaviour data are collected as input sources. These logs are processed using AWS Lambda and analyzed by Amazon Bedrock AI to detect suspicious login activities and abnormal behaviour patterns. Based on the AI decision, the system sends real-time alerts through SNS, updates the security dashboard with warning messages, and securely stores activity logs in Amazon S3 for future auditing and analysis. The frontend dashboard is built using React and the backend is developed with Flask and Python using AWS boto3 integration.

In Scope

Real-time AWS login and activity monitoring, CloudTrail and GuardDuty log analysis, AI-based suspicious activity detection, user behaviour analysis, automated threat classification, instant SNS/email alerts, live security dashboard monitoring, suspicious login tracking with IP and location details, secure activity log storage in Amazon S3, role-based user authentication, security audit reporting, and AI-powered cloud security insights for AWS environments.

Out of Scope

Support for non-AWS cloud platforms such as Google Cloud Platform (GCP) and Microsoft Azure, Kubernetes cluster security monitoring, enterprise single sign-on (SSO) integration, mobile application support, advanced machine learning model training, automatic threat remediation using SOAR platforms, real-time WebSocket-based event streaming, blockchain-based security logging, multi-cloud hybrid infrastructure management, and offline security monitoring are outside the current scope of CloudShield AI.

Future Enhancements

- ◆ Add Dark Mode support for the security dashboard to improve user experience and accessibility
- ◆ Implement Security Health Score Cards to display overall AWS account security status
- ◆ Introduce AI-based threat severity visualization with interactive charts
- ◆ Integrate advanced AI models for better anomaly and threat prediction
- ◆ Add live notification panels for real-time security alerts

Conclusion

CloudShield AI is a well-designed AI-powered AWS security monitoring system that demonstrates strong cloud security integration and real-time threat detection capabilities using AWS services such as CloudTrail, GuardDuty, Lambda, SNS, and Amazon Bedrock. The project effectively provides suspicious login detection, user behavior analysis, automated security alerts, and centralized monitoring through an interactive dashboard. Its primary future challenges include improving scalability for handling large-scale AWS activity logs, enhancing AI threat detection accuracy, strengthening credential and access security mechanisms, and expanding support for advanced automated incident response and multi-cloud security environments.

Project Type: AI-powered SaaS Web Application for AWS Cloud Security Monitoring and Threat Detection